

STATISTICAL MACHINE LEARNING (WS2025/26)

HOMEWORK: CONFIDENCE INTERVAL

Assignment (2 points) Assume a predictor $h: \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{Y} = \{0, 1, \dots, Y - 1\}$, is applied on i.i.d. sample $((x_i, y_i) \in \mathcal{X} \times \mathcal{Y} \mid i = 1, \dots, n)$.

Given:

- A sequence of class predictions $(h(x_1), \dots, h(x_n))$,
- A sequence of true class labels $(y_1, \dots, y_n)$,
- A loss function $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$,
- An error level $\delta \in [0, 1]$,

Your task is to compute a $(1 - \delta)$ -confidence interval (R_L, R_R) such that

$$R(p, h) \in (R_L, R_R)$$

with probability at least $(1 - \delta)$. Here,

$$R(p, h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$$

denotes the true (expected) risk of the predictor.

You have to fill in code for the following Python function:

```
def confidence_interval(true_y, pred_y, loss, delta ):
    # Input:
    # true_y is np.array(1,dtype=int) representing the true class labels;
    # a label is an integer from 0 to Y-1
    # pred_y is np.array(1,dtype=int) representing the predicted class labels;
    # a label is an integer from 0 to Y-1
    # Loss is np.array((Y,Y)) whose Loss[y,yy] represents the loss
    # incurred when the true label is y and prediction is yy
    # delta is a scalar from (0,1) representing the probability of failure
    #
    # Output:
    # R_L, R_R are scalars such that R_L<=R_R

    return R_L, R_R
```